Student Absenteeism Early Warning System

Chicago Public Schools District | Executive Summary

The Problem

Every year, thousands of students become chronically absent — missing more than 10% of school days. By the time administrators notice, it is often too late. These students fall behind academically, are less likely to graduate, and face long-term consequences.

Today, there is no reliable way to identify which students will become chronically absent *before* it happens. Staff are reacting after the damage is done, rather than preventing it.

Our Solution

We built an early warning system using a **Random Forest model** — a proven approach that learns patterns from **400,000+ historical student records** to score every enrolled student's risk at the start of each year.

It uses data already in district systems (age, grade, school, housing stability, special education status). No new data collection needed. A live interactive dashboard lets staff view scores and download action reports.

Current Findings

Built by 1 data scientist in 2 weeks:

Metric	What It Means	Result
Overall Accuracy	Students correctly classified per 100	**~75–80%**
Precision	When we flag a student, how often we're right	**~70–75%**
Recall	Of all at-risk students, how many we catch	**~75–80%**
Confidence (AUC)	Ability to rank risk correctly	**~80–85%**

In plain terms: The model catches ~3 out of 4 students who will become chronically absent — before it happens. Top risk drivers: student age, housing instability, school location, special education status.

Business Impact

For our ~313,000 active students:

- **Proactive intervention:** Reach at-risk families in September, not February. Research shows 15–30% reduction in absenteeism with early-warning systems.
- **Graduation outcomes:** Chronic absenteeism is the #1 predictor of dropping out. Every student kept on track is one more graduate.
- **Dollar savings:** Preventing just 50 students from dropping off track = **$13M+** in long-term community value.
- **Resource focus:** Target support at the specific schools, ages, and populations with highest risk.
- **Equity:** Every student gets attention based on data-driven need, not just who is most visible.

Path Forward

With further investment beyond the initial 2-week prototype:

- **More data** (prior attendance, grades, engagement) → accuracy above 90%
- **Real-time updates** → weekly score refresh catches students as patterns shift
- **District-wide scale** → every school with tailored recommendations
- **Measurable ROI** → track exactly how many students stayed on track due to intervention

Assumptions Behind Dollar Estimates

1. Lifetime cost of one dropout ≈ $272,000 (lost earnings + public services). *Source: Alliance for Excellent Education*
2. Chronically absent students are 5–7x more likely to drop out. *Source: U.S. Dept. of Education*
3. Conservative 10–15% intervention success rate applied to ~400–500 highest-risk students ≈ 50 students saved. *Source: Chicago Consortium on School Research*
4. Figures are long-term community value over a lifetime, not immediate annual budget savings.
5. 15–30% absenteeism reduction from early-warning systems; we use the lower end. *Source: Everyone Graduates Center, Johns Hopkins University*

Bottom line: A modest investment can keep hundreds more students in school, improve graduation rates across the district, and deliver millions in long-term community value.

For a live demonstration, visit the interactive dashboard to explore risk scores, adjust sensitivity, and look up individual students.